

Large Eddy Simulation of high-Reynolds number flow past a rotating cylinder

S.J. Karabelas *

National Technical University of Athens (NTUA), Department of Chemical Engineering, Computational Fluid Dynamics Unit, 157 80 Athens, Greece

ARTICLE INFO

Article history:

Received 13 May 2009

Received in revised form 15 January 2010

Accepted 9 February 2010

Available online 27 April 2010

Keywords:

Magnus effect
Rotating cylinder
Load stability
Turbulence

ABSTRACT

In the present study, uniform flow past a rotating cylinder at $Re = 140,000$ is computed based on Large Eddy Simulation (LES). The cylinder rotates with different spin ratios varying from $a = 0$ to $a = 2$, where a is defined as the ratio of the cylinder's circumferential speed to the free-stream speed. The Smagorinsky model is applied to resolve the residual stresses. The present commercial code is validated based on available numerical and experimental data. The results agreed fairly well with these data for the cases of the flow over a stationary and over a rotating cylinder. As the spin ratio increases, the mean drag decreases and the mean cross-stream force acting to the cylinder increases. The vortices (time-averaged) downstream of the cylinder are displaced and deformed and the vortex that is close to the region of the fluid's acceleration shrinks and eventually collapses. By increasing a , the flow is also stabilized. It is observed that the vortex shedding process is suppressed. Specifically, the flow is unstable in load terms for spin ratios up to 1.3. After this critical value, the flow is transitional for a few dimensionless time units demonstrating the well-known von-Karman vortex street and then it becomes stable with almost constant loads. An encouraging outcome resulting from this study is that the LES computations could be accurate for high- Re sub-critical flows with grids of medium resolution combined with a validated sub-grid scale model and a low-diffusive discretization scheme.

© 2010 Elsevier Inc. All rights reserved.

1. Introduction

Bluff body flows have been the target of study for many scientists, since the physics of these flows is very complex and they require special attention to their modelling and numerical solution. They are also ideal cases for the validation of different approaches to turbulence modelling. One of these is Large Eddy Simulation (LES), a promising technique that has recently started to gain popularity even for high- Re flows. However, it is well known that for high- Re regimes the sub-grid-scale and the near-wall modelling become crucial for the accuracy of the computations. In cylinder flows, the results are not very promising for super-critical Re numbers, at least based on the preliminary study by Catalano et al. (2003) who assessed the validity of LES with near-wall modelling for a flow past a circular cylinder up to $Re = 2 \times 10^6$. Their results departed significantly from those reported in experiments. Nevertheless, they stated that their study is at a preliminary stage and no well-justified conclusions could be drawn. In contrast, the numerical results for sub-critical but still high- Re flows past a cylinder are satisfactory. Breuer (1999, 2000) conducted relevant studies at $Re = 140,000$, where he found sufficient agreement with the well-organized experiment of Cantwell and Coles (1983). Elmiligui et al. (2004) studied the same problem at $Re = 50,000$ and $Re = 140,000$, but the free-stream flow was not laminar. Medium turbulence intensity was imposed on the inlet of

the domain. The results were quite different from those of Breuer (2000) in terms of the drag coefficient and Strouhal number, but they agreed well with the numerical experiments of Travin et al. (2000) and Hansen and Forsythe (2003), who used relevant boundary and initial conditions.

The above studies cited in the open literature refer to the modelling of the flow past a stationary cylinder for high- Re numbers. The more challenging case of the flow past a rotating cylinder has not yet been investigated extensively. In the laminar regimes two highlighted studies are those of Mittal and Kumar (2003) and Padrino and Joseph (2006). The Re number based on the diameter of the cylinder and the free-stream flow is 200 in the first study and 200, 400 and 1000 in the latter study. In the context of these articles, the physics of the laminar flow past a spinning cylinder is analysed for various spin ratios whereas the lift, drag and pressure coefficients are computed. The well-known Magnus effect is examined in this framework and the results (for the lift force) are compared with those of potential flow (Zdravkovich 1997). An interesting study of Stojkovic et al. (2002) has been also found in the literature, in which high rotation rate effects to the mean loads are investigated. The values of a varied from 0 to 12 and the characteristic Re number based on the diameter of the cylinder was 100. Many additional results have been obtained, including the frequencies of the wake instability and the distinct changes of the flow structure. The load stability of laminar flows has also received considerable attention. It has been generally found in the preceding studies that the rotational effects suppress the vortex

* Tel.: +306 977207390.

E-mail addresses: stkarabelas@gmail.com, stkarabelas@yahoo.gr

shedding activity at a specific range of spin ratio. More specifically at $Re = 200$ it is universally accepted that for $(1.91 \leq a \leq 4.4) \cup a > 4.8$, where a is defined as $a = \frac{U_\theta}{U_\infty}$ with U_θ being the circumferential velocity and U_∞ the free-stream velocity, the flow becomes steady (Mittal and Kumar (2003)). Analogous conclusions have been reached in the detailed study of Stojkovic et al. (2003). They confirmed the existence of the second vortex shedding mode for the entire Reynolds number range $60 \leq Re \leq 200$ and a complete bifurcation diagram $a(Re)$ was presented. A well-organized numerical study was also published by Ingham (1983). In that paper, the Navier–Stokes equations were solved via the finite difference method in order to examine the asymmetrical flow in uniform viscous liquid at Re numbers 5 and 20 and dimensionless ratio a from 0 to 0.5. Kang et al. (1999) also contributed significant work to this research area. Sequential numerical simulations at Re equal to 40, 60, 100 and 160 in the range of $0 \leq \alpha \leq 2.5$ were performed. They observed that at $60 \leq Re \leq 160$ the maximum value of a , which favours flow instability varies logarithmically when plotted against the Re number.

In transitional or fully turbulent flows analogous conclusions have not been reached, at least, to the author's knowledge. There are two research articles that examine rotating cylinder flows at high- Re numbers and spin ratios up to 1. These are the studies of Aoki and Ito (2001) and Elmiligui et al. (2004). For values greater than 1, the present study might be the first attempt to resolve the flow phenomena.

The objective of the present research work is to resolve the physics of the flow past a spinning cylinder for ratios up to 2 at $Re = 140,000$. Besides the computation of the integral coefficients and the statistics of turbulence, load stability has been studied for the above range of a and a new 'limit' for wake stability has been established.

2. Computational grid and numerical modelling

The 3-D incompressible time-dependent viscous Navier–Stokes equations are solved via the finite-volume code FLUENT 6.3, which is compatible with multi-block structured grids. The computations are performed in a single computer without any parallelization or

domain decomposition. The equations are discretized in space by the low-diffusive central differencing scheme of second order accuracy. Temporal discretization is fully implicit and second order accurate for all the present computations. Within every time step, the Poisson equation for the pressure correction is formulated by the PISO velocity–pressure coupling scheme and is solved by a point implicit Gauss–Seidel iteration, which is accelerated by an algebraic multi-grid method (AMG). Sub-grid scale modelling is based on the Smagorinsky model (Smagorinsky, 1963) with a 'wall-sensitive' length-scale defined as $l_s = \min(kd, C_s V^{1/3})$, where k is the von-Karman constant, d is the distance from the wall, C_s the Smagorinsky constant (equal to 0.1 in the present study) and V the volume of the cell.

The dimensionless time step $dt' = dt \frac{U_\infty}{D}$ is chosen to be 10^{-3} , where U_∞ is the free-stream velocity and D the diameter of the cylinder. This value fulfils the accuracy requirement (and stability requirement for explicit solvers) $CFL \leq 1$ and it is also chosen based on the values cited in relevant research articles (Breuer, 2000; Catalano et al., 2003). The grid size is quite thin close to the cylinder and it is capable of resolving most of the developed boundary layer, thus any wall functions did not apply ($y^+ < 5$ at all spin ratios).

A single curvilinear (body-fitted) O-type grid (Fig. 1) based on the study of Padrino and Joseph (2006) is used for all the computations. In the cross-sectional plane 125×125 grid points are used, while in the spanwise direction 32 points are distributed uniformly. The entire domain has a radial extension of $L = 20D$ which appears to be a proper choice according to the preliminary computations of Breuer (1998). The spanwise extension was carefully chosen to be $Z = D$ based on the study of Breuer (1999) who assessed the influence of the spanwise extension on the LES computations of the flow past a circular cylinder at $Re = 140,000$. Although the spanwise length of one cylinder diameter may not be capable of resolving all the turbulent structures, it is revealed in the latter study that the results for $Z = D$ agree fairly well with relevant experimental data and on some occasions the results match even better with experiments than for $Z = 2D$. In the present study this is verified below. Moreover, the computational time needed to converge statistically the results is at least four times

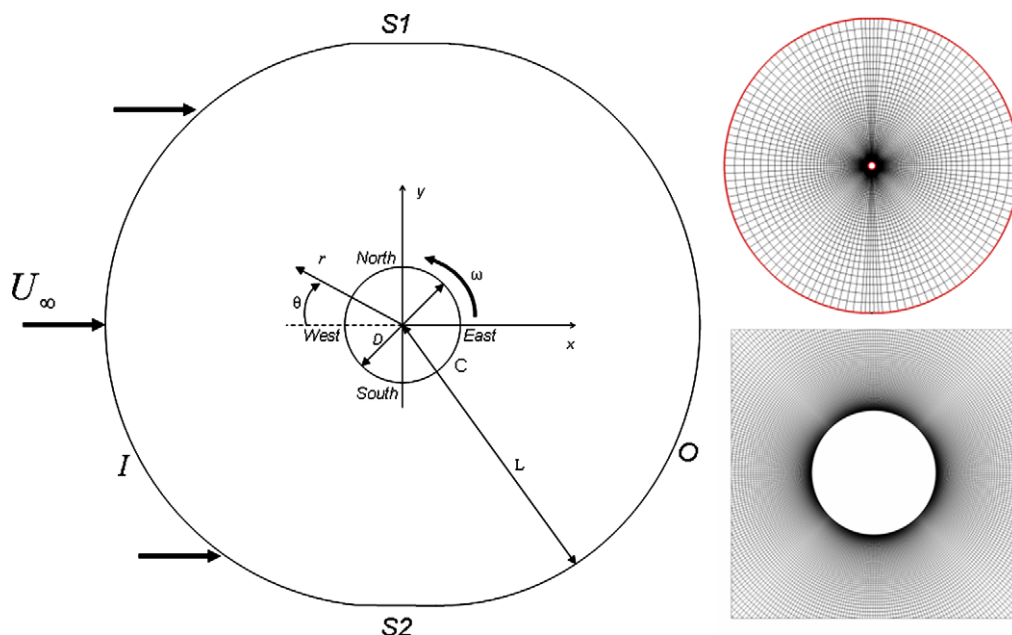


Fig. 1. Computational domain (side section) and the system of reference for the numerical simulations. The azimuth angle θ is measured in a clockwise way while the cylinder rotates in the counter-clockwise direction. A panoramic and a close-up view of the structured grid adopted are also presented.

less than using a grid with double spanwise length with cells of the same aspect ratio (non-elongated cells). However, it has to be mentioned that based on the studies of Breuer (1999, 2000), this value of the spanwise length reduces the drag coefficient roughly 20% and increases the separation angle by 1–3°.

3. Description of the test cases and validation of the code

The present study investigates the rotational effects induced by the no-slip condition on the cylinder's surface at $Re = 140,000$. The choice of this number was intentional, because at this value of Re , experimental and numerical data were available from the literature. The spin ratio of the cylinder $a = \frac{U_\theta}{U_\infty}$, varies from 0 to 2. Close to the value, which was firstly observed to produce load stability, it was necessary to perform more runs in order to indicate as precisely as possible the 'limit' where the wake instability is strongly suppressed.

Two symmetry boundary conditions are applied far away from the cylinder (S1: $85.8^\circ \leq \theta \leq 94.2^\circ$) and (S2: $265.8^\circ \leq \theta \leq 274.2^\circ$) as shown in Fig. 1, a velocity inlet condition 'free of turbulence intensity' is prescribed to the region on the left (I : $-85.8^\circ \leq \theta \leq 85.8^\circ$) of the domain and a zero stream-wise gradient 'outflow' condition to the region on the right (O : $94.2^\circ \leq \theta \leq 265.8^\circ$). Periodicity of the flow is assumed in the spanwise direction. The cylinder rotates in the counter-clockwise direction, thus the no-slip condition $\vec{U} = (\frac{aU_\infty}{R})\vec{z} \times \vec{r}(x, y)$ is imposed on the cylinder's surface (C).

The present study is concerned with a high- Re regime where transition to turbulence occurs inside the thin shear layer just after the line of separation. Since available data exist for the non-rotating case, it is considered necessary to validate the implemented code. This is carefully done by comparing the present results with those of Breuer (2000) and the measurements of Cantwell and Coles (1983). At this Re number the flow is turbulent and statistically non-stationary. The fluid accelerates up to the middle of the cylinder and then the laminar boundary layer separates. After a short time the shear layer becomes turbulent and, on each side of the cylinder, both separated shear layers are quite unstable forming the commonly-observed vortex shedding pattern downstream. It appears from the cited studies and the present results that the flow is statistically symmetric. Table 1 summarizes the results of a collection of simulations by Breuer (2000), the measurements from the experiments of Wieselsberger et al. (1923) and the measurements of Cantwell and Coles (1983). It has to be acknowledged that in the experiments there are several parameters which significantly influence the measurements both locally and globally. The most important are the blockage ratio, the aspect ratio of the cylinder, the turbulence intensity of the free-stream flow (which is always non zero for all the experiments), the end conditions, the roughness of the model, the accuracy of any implemented virtual balance devices or hot-wire anemometry equipment and the precision of the calibration model used for the computation of the loads. It becomes obvious that differences arise between the set-up of the numerical modelling and the experimental conditions, even for the most carefully prepared experiment. Neverthe-

less, a quite clear picture for the evaluation of the present numerical results could be drawn from the cited experiments.

From Table 1, it may be concluded that the numerical and experimental results are in good agreement. The Strouhal number converges to 0.2 except for the experiment of Cantwell, which is under-predicted. The back-pressure coefficient and the angle of separation agree fairly well in the numerical simulations. However, in the experiment the reported angle is measured with respect to the appearance of the inflexion point and it is lower than the separation angle, thus a direct comparison could not be made. The drag coefficient is close to the other numerical computations but a difference is observed with the experiments.

A more qualitative evaluation of the present results may be obtained by comparing them with other data in terms of the time-averaged and spanwise-averaged velocity fields. Fig. 2 provides the dimensionless stream-wise velocity profile at the symmetry line $y = 0$. The present results appear to over-predict the length of the recirculation area and the magnitude of the reversed velocity with respect to the cited results of Breuer (2000). In the near-middle and far wake, the present computations agree quite well with the experimental measurements. Unfortunately, no experimental evidence exists for the velocity profile in the vicinity of the cylinder. In Fig. 3, the computed dimensionless normal velocity V at a constant x -position $x = D$ in the near wake almost coincides with the numerical data of Breuer (2000) and is slightly under-predicted compared to the measurements of Cantwell and Coles (1983).

Second-order moments have been also investigated, including the variance of the normal fluctuations $\langle vv \rangle$ and the Reynolds shear stress $\langle uv \rangle$. Before this topic is discussed, some theoretical remarks should be made. The flow over a cylinder at sub-critical regimes is strongly unsteady due to the strong vortex – shedding which occurs behind the cylinder. The vortex shedding is accompanied by turbulent fluctuations and, therefore, the computed flow

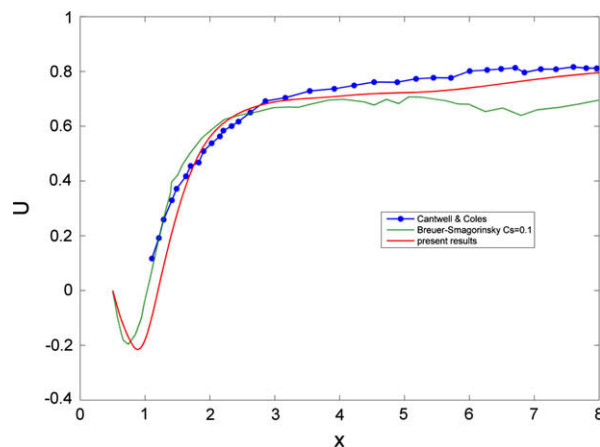


Fig. 2. Time-averaged dimensionless stream-wise velocity along the centreline $y = 0$: Breuer's results, present results and the experimental measurements of Cantwell and Coles (1983).

Table 1
Experimental and numerical results in the literature compared to the present ones.

	Grid	SGS model	C_d	C_{peak}	St	$\theta_{sep.} (^\circ)$
Present	$125 \times 125 \times 32^*$	Smag, $C_s = 0.1$	1.03	−1.19	0.22	96.90
Breuer (2000)	$165 \times 165 \times 64^*$	Smag, $C_s = 0.1$	0.97	−1.08	0.23	96.60
Breuer (2000)	$325 \times 325 \times 64^*$	Smag, $C_s = 0.1$	1.05	−1.22	0.19	93.62
Cantwell and Coles (1983)	(exp.)	–	1.23	−1.21	0.18	>77**
Wieselsberger et al. (1923)	(exp.)	–	≈ 1.20		≈ 0.2	

* All computations have been done for spanwise length $Z = D$.

** Angle corresponds to the appearance of the inflexion point.

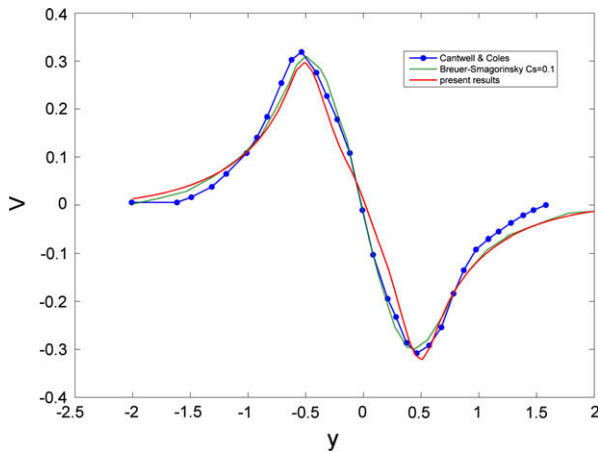


Fig. 3. Time-averaged dimensionless normal velocity at $x = D$: Breuer's results, present results and the experimental measurements of Cantwell and Coles (1983).

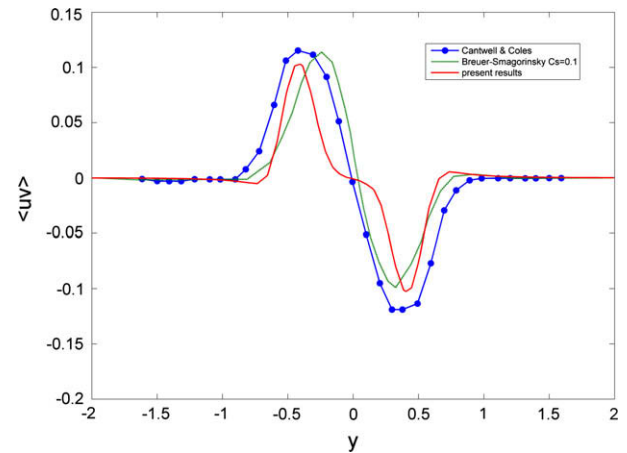


Fig. 5. Total resolved dimensionless shear stress at $x = D$: Breuer's results, present results and the experimental measurements of Cantwell and Coles (1983).

variables are often decomposed into three instead of the traditional two components: the time-averaged mean value, the periodic component and the turbulent fluctuation. In order to separate the effects of the two latter components, phase-averaging is necessary. However, in the present study, no phase-averaging has been performed, therefore, the Reynolds stresses are composed of the 'periodic' and the 'turbulent' component.

In Fig. 4, the dimensionless mean-square fluctuation of the normal velocity, denoted $\langle vv \rangle$ is plotted at $y = 0$. The present LES results appear to converge better than Breuer's (2000) computations with the measurements. Close to the cylinder's surface there is a large difference between the numerical data and the measured value regarding the location of the peak value of $\langle vv \rangle$. From this figure it is seen that the cited numerical data overestimate $\langle vv \rangle$ and the peak value is found to be farther from the rear of the cylinder. In the near and far wake the present results agree well with the measurements. An astonishing conclusion reported in Breuer's (2000) study was that grid refinement did not lead to closer agreement between the numerical results with the measurements and in some cases the differences became greater. This was also the case when the second-order moment $\langle uv \rangle$ is plotted at $x = D$, Fig. 5. The present results agree well with respect to the maximum values and their positions obtained from the measurements. Nevertheless, they are distributed differently over the suction side.

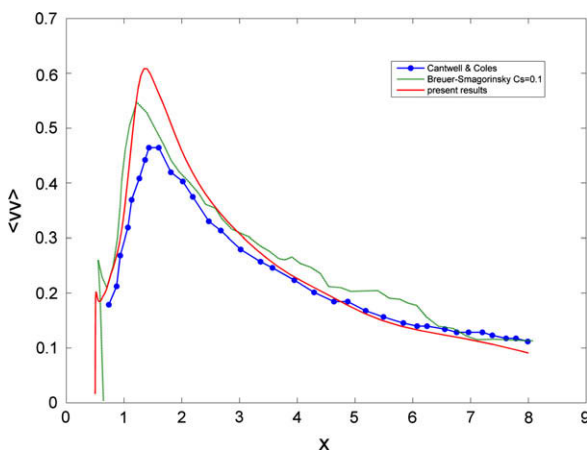


Fig. 4. Mean dimensionless square fluctuations of the normal velocity at $y = 0$: Breuer's results, present results and the experimental measurements of Cantwell and Coles (1983).

The other numerical results over-predict the turbulent shear stress but they are distributed more consistently. It is also confirmed for the latter numerical computations that their results for the finer grid deviate more from the experimental measurements of Cantwell and Coles (1983). Explanations for this unexpected trend are offered in the same study of Breuer (2000).

The deviations between the present numerical data and Breuer's data are attributed mainly to the different near-wall grid resolution, the use of a different damping function for the Smagorinsky length-scale and, possibly, slightly to the different cross-sectional area. The differences with the experimental data are observed for several reasons. In the context of a numerical simulation it is not possible to simulate exactly the experimental conditions. In the present case, there are many major differences between the set-up of the numerical and the experimental process. The most important are the different spanwise length of the cylinder and the different turbulence level at the inflow (see Cantwell and Coles, 1983; Breuer, 2000). The above parameters suffice to produce discrepancies between the numerical and the experimental results.

4. Results and discussion

4.1. Basic flow physics

The physics of the turbulent flow over a rotating cylinder is quite different from that over a stationary cylinder. It is not always more complex as it might be expected. Before proceeding to the quantitative results of this study it is convenient to present the general flow features of rotating cylinder flows at $Re = 140,000$. Initially, five spin ratios were investigated and then three additional runs were performed to examine the flow stability. In Fig. 6, the streamlines of the time-averaged and spanwise-averaged velocity fields are displayed. For the non-rotating case, in accordance with the experimental data, the computations predict an attached recirculation region downstream of the cylinder. The separation angle is found to be 96.9° , thus the flow separates after the apex of the cylinder. Two smaller counter-rotating vortices are observed on the downstream side of the cylinder, in contrast to the results of Breuer (2000). A possible explanation for this difference could be the implementation of a different length-scale (close to the walls) in the Smagorinsky model (which was the Van Driest damping function in the latter study). In an attempt to identify the cause of the above difference, an extra run at $Re = 140,000$ and at $a = 0$ was performed using the dynamic model, Germano et al. (1991). The

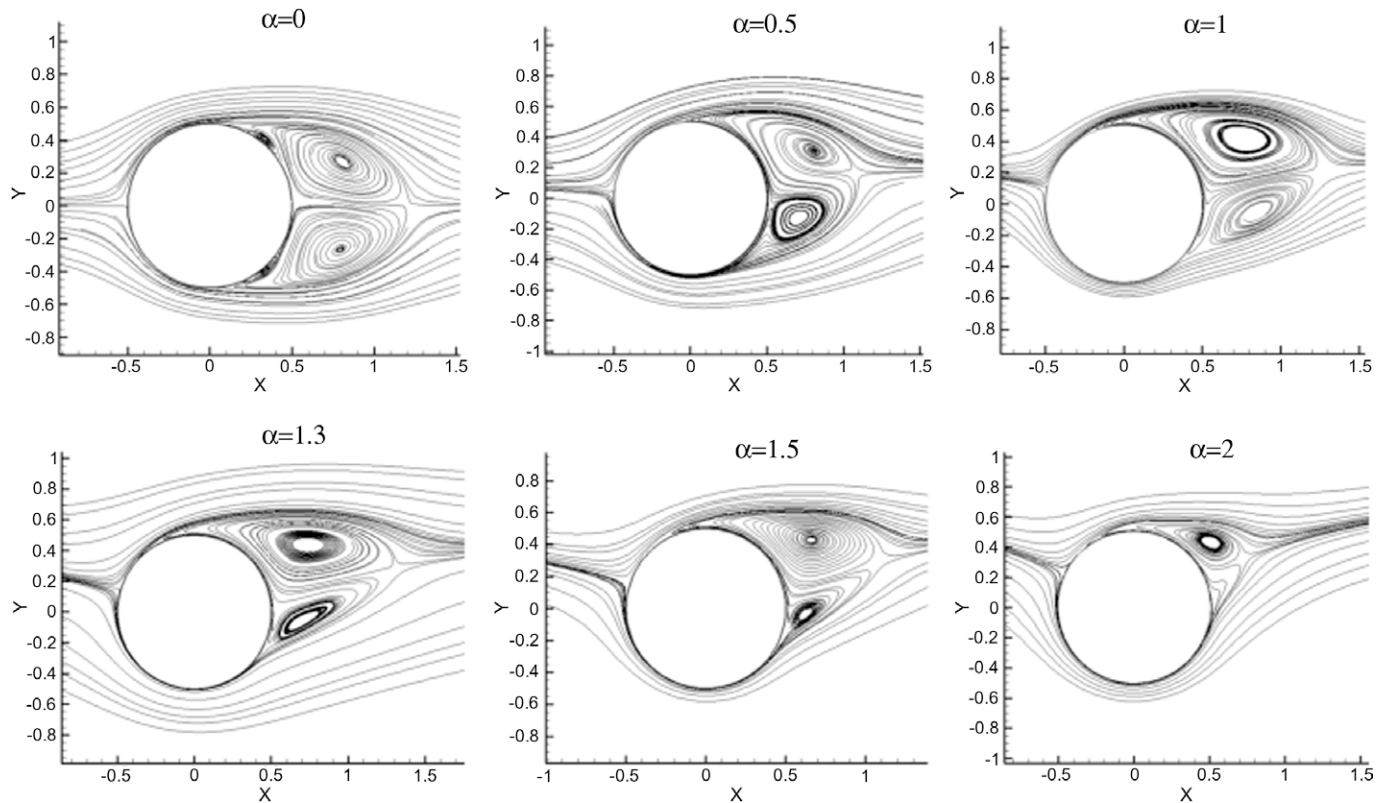


Fig. 6. Streamlines of time-averaged velocity fields for spin ratios ranging from $a = 0$ to $a = 2$.

streamline results are found to be almost identical with those ones of the classical Smagorinsky model, thus the use of a different damping function does not explain the different flow patterns. Based also on the fact that the solution method is the same and both grids are quite similar, a convincing explanation could not be given.

Different patterns are observed when the cylinder rotates (in the counter-clockwise direction). Generally speaking, the rotating fluid driven by the motion of the cylinder is superimposed on the free-stream flow on the lower surface ($260^\circ < \theta < 340^\circ$). This naturally leads to a global increase of the fluid's velocity. In the region ($20^\circ < \theta < 100^\circ$) the rotating fluid opposes the free-stream flow. This effect retards the outer flow which actually slides over fluid moving in the opposite direction and, as will be shown this triggers the deflection of the outer flow towards the transverse direction and allows the rotating boundary layer (RBL) to dominate the downstream side of the cylinder. For $a = 0.5$, the no-slip condition does not induce large circumferential velocities in the vicinity of the cylinder. Over the front and upper part the RBL develops only in the viscous sub-layer. Correspondingly, the pair of vortices downstream of the cylinder is asymmetric. The lower vortex contracts while the upper one is little enlarged. In the south area the RBL energizes the free-stream flow which remains attached to the cylinder, thus no separation occurs on the lower side (the low pressure side) of the cylinder and the vortex is formed at a considerable distance from the wall. Over the upper region of the cylinder the gradual development of the RBL causes the bulk of the outer flow to move transversely and it consequently forms the upper vortex structure. On the rear side the RBL develops free with the slight effect of the recirculating fluid which had been previously deflected. Similar conclusions may be drawn for $a = 1$. The circumferential velocities in the viscous sub-layer are comparable with the magnitude of the free-stream velocity and the effects are stronger than in the previous case. The outer flow quickly moves transversely, allowing the formation of a large upper vortex.

The bottom vortex is slightly larger than in the previous case. The system of both vortices is asymmetric but it is also rotated counter-clockwise with respect to the x (longitudinal) axis. The front stagnation point is evidently displaced by approximately at $\theta = 10^\circ$.

As the spin ratio further increases, the bottom vortex shrinks and the upper expands. The front stagnation point moves azimuthally to greater θ and the flow beneath the lower surface of the cylinder approaches the classical flow induced by pure rotation. The case $a = 2$ is of particular interest. It is clear that the RBL is significantly stronger than the mean magnitude of the free-stream velocity. The stagnation point is displaced to $\theta = 28^\circ$ and is displaced away from the wall. Beyond this value of the azimuthal angle, the outer flow slides over a quite thick fluid layer which moves in the opposite direction. The resulting upper vortex is not as large as expected, but it is slenderer than in the previous cases. The lower vortex has collapsed. A possible explanation for this, might be the combined action of the position of the upper vortex and the high momentum of the fluid coming from the lower side of the cylinder.

4.2. Mean loads and higher order moments

Rotational effects greatly modify the loads exerted by the fluid on the cylinder. Lift and drag coefficients are plotted for various spin rates in Fig. 7 and compared with the computations of Aoki and Ito (2001) and some available experimental data (published in the same study). Also some laminar data at $Re = 40$ and $Re = 160$ have been selected from the study of Kang et al. (1999) and the results of the potential theory are also presented in the same figure for further comparison and discussion.

The lift force is generated by the acceleration of the fluid beneath the cylinder and its deceleration above it (for counter-clockwise rotation). It is seen that the lift increases linearly with a for low spin ratios, whereas for higher values it increases more steeply. The results agree quite well with Aoki's computations,

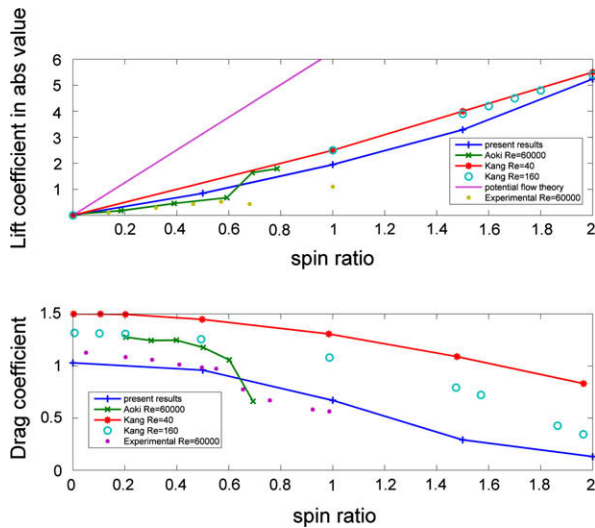


Fig. 7. Lift and drag coefficient versus spin ratio: Aoki and Ito (2001) results at $Re = 60,000$, present results at $Re = 140,000$ and the experimental measurements of Aoki and Ito (2001) at $Re = 60,000$.

although the Re number is not the same. The drag coefficient reduces as a increases because the upstream stagnation point is shifted azimuthally from the longitudinal direction (x -axis) and the magnitude of the pressure drag, which is the main contributor to the drag coefficient, decreases. This reduction in drag is counter-balanced by an increase in lift. The results for the mean drag coefficient are quite satisfactory when compared with the experimental measurements. Differences did not exceed 10% of the nominal values. The under-estimation of the mean drag coefficient is attributed to the modest spanwise length of the domain (one diameter).

Interesting conclusions may be drawn from the comparing between the laminar and turbulent cases, which is examined in this figure. It is clear from Fig. 7 that the spin ratio generally affects the forces applied to the cylinder in a similar manner, no matter whether the flow is turbulent or laminar. However, some differences are apparent. It appears that the lift coefficient in the laminar case increases at a comparable rate on average as for turbulent flow, but the value of C_L in laminar is higher. It is also observed that for high a the differences between the laminar and turbulent values of C_L and C_D diminish. The drag coefficient in the laminar regime is greater for all a and its reduction is steeper at $Re = 160$. The basic reason for these differences is that the laminar patterns are quite different from the turbulent ones. In laminar flows, the stagnation point usually lies further from the cylinder's surface and is located at different circumferential locations positions than in the turbulent regime, thus the static pressure distribution is greatly different. Other important factors could be the high separation angle and the increasing friction drag, when turbulence is present.

In Fig. 8, the mean pressure coefficient is plotted for various spin ratios. The reference pressure p_∞ satisfies the condition $C_p = 1$ at $\theta = 0^\circ$ and $a = 0$, while for higher values of a the stagnation point moves accordingly. The reduction of the static pressure on the lower side is clearly evident where the pressure coefficient exceeds the value of -6 (the azimuthal angle is measured anti-clockwise rather than the convention indicated in Fig. 1). The position of the minimum C_p is displaced towards the lower surface as the spin ratio increases. Similarly, the position of the maximum C_p value moves towards the upper surface as a increases. The pressure coefficient peaks at the stagnation point, which can reach significantly higher values (in terms of the azimuthal angle) than the zero one at $a = 0$; thus, the maximum value of the azimuthal angle is approximately $\theta = 28^\circ$ at $a = 2$ (in the diagram the angle is 332°).

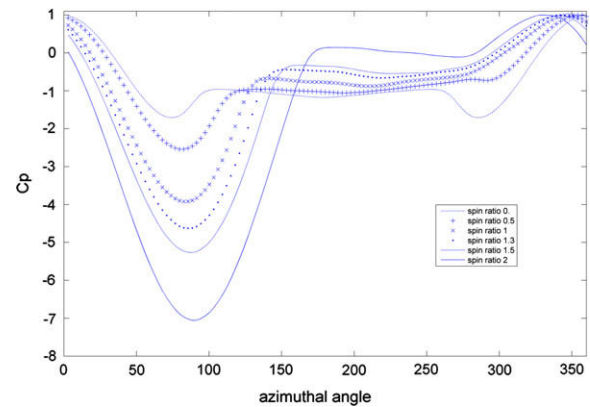


Fig. 8. Pressure coefficient C_p computed along the cylinder's surface for different spin ratios. For all the cases the convention $C_p = 1$ at $\theta = 0^\circ$ is adopted.

For a more detailed investigation of the turbulence field developed, second-order moments are examined. As has already been noted, these moments are the sum of the periodic vortex shedding (low-frequency) oscillations plus the random turbulent (high-frequency) fluctuations. In Fig. 9, the total resolved kinetic energy of the fluctuations $k_f = 0.5(\langle u^2 \rangle + \langle v^2 \rangle + \langle w^2 \rangle)$ is presented. Strictly speaking k_f is equal to the turbulent kinetic energy. However, the turbulent kinetic energy is commonly understood to be the product of the turbulent fluctuations and excluding any other oscillations, which are produced by wake instabilities. Thus, the definition above is used specifically to distinguish the turbulent kinetic energy (energy produced by high-frequency disturbances) from the total resolved kinetic energy of fluctuations k_f .

In the case of the stationary cylinder, the energy reaches a maximum downstream of the cylinder at $x = 1.3D$. The major fraction of this energy is attributed to the strong vortex shedding oscillations rather than the turbulent fluctuations. In the same plot, the point of transition to turbulence appears to lie downstream of the separation point. For greater spin ratios, the region of maximum energy is shifted upwards forming an angle with the longitudinal axis. For $a = 0.5$ and $a = 1.0$, the flow becomes turbulent both at lower azimuthal angles than $\theta = 90^\circ$ and close to the point where the free-stream flow is deflected from the surface of the cylinder (see Fig. 6). The values of k_f at these azimuthal angles are computed solely on the turbulent fluctuations since there is no contribution from vortex shedding there. Careful inspection shows that for spin ratios greater than one the downstream region attached to the cylinder is entirely turbulent. In contrast, for lower a , laminar areas could be observed.

The case of the highest spin ratio, $a = 2$, considered in more detail. The region of maximum k_f is close to the upper surface. As will be seen later, at this rotation rate the vortex shedding is strongly suppressed after a short dimensionless time. Therefore, the total resolved kinetic energy of the fluctuations is composed mainly of the energy derived from the turbulent fluctuations. On this occasion, the flow appears to be highly turbulent close to the cylinder's surface. On the other hand, for lower ratios the contribution of the periodic oscillations to k_f prevails. It also appears from this figure that the total resolved kinetic energy of the fluctuations decreases significantly with a . However, the turbulent kinetic energy increases with the spin ratio since higher vorticity values are observed close to the upper surface. The explanation for this result is closely related to the stability of these flows. As will be seen in the next section, the vortex shedding is gradually suppressed and it is possible that for high values of a it would be diminished. The effects are straightforward for the distribution of the kinetic energy. As a increases, the periodic oscillations are damped and

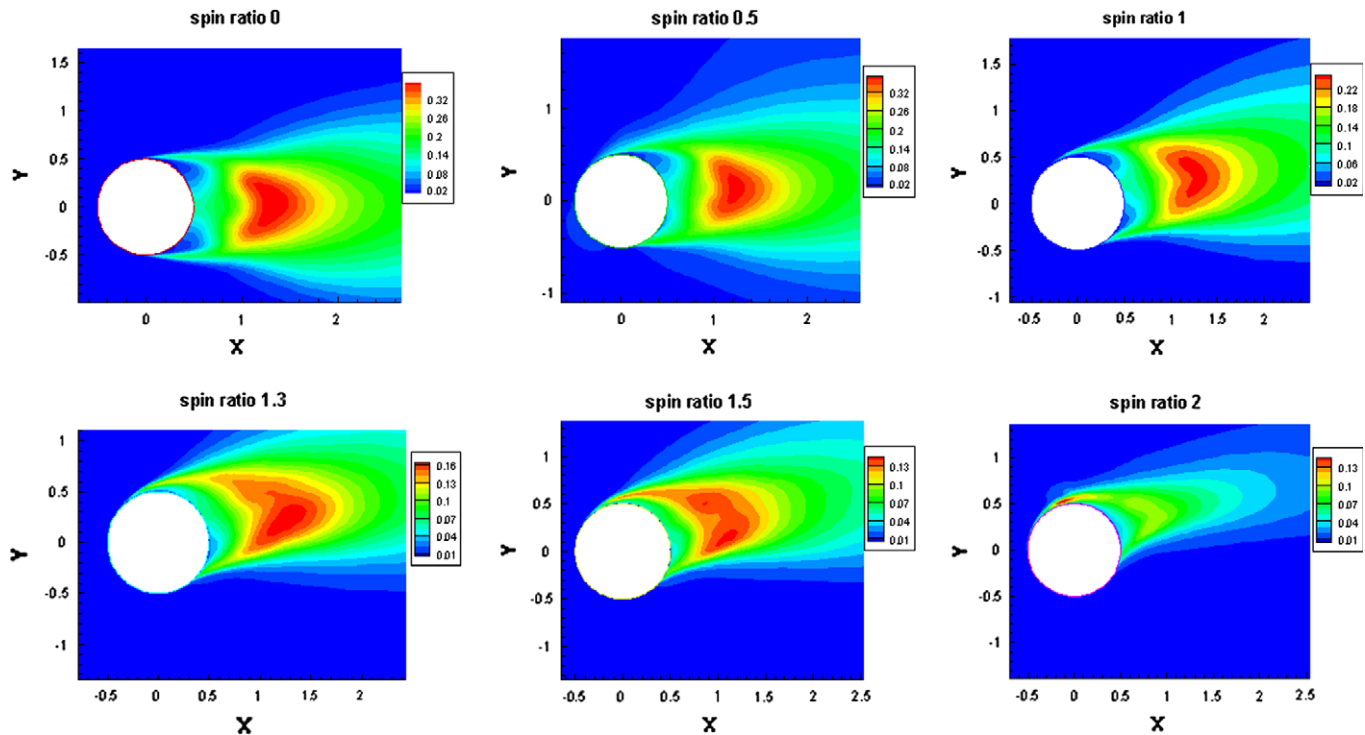


Fig. 9. Contours of the total resolved kinetic energy of the fluctuations k_f plotted in the near wake for the examined spin ratios.

the turbulent fluctuations are intensified. Based on the above scenario three different cases may be deduced: (1) for low a , the major source of k_f is the periodic oscillations, (2) for medium a , the contributions of the periodic and turbulent fluctuations are equally important and (3) for high a the turbulent fluctuations become the main source of the total resolved kinetic energy of the fluctuations. Thus, the reduction of k_f is due to the damping of the periodic oscillations.

Reynolds shear stress $\langle uv \rangle$ is plotted in Fig. 10 in the near wake. Similarly to the case of k_f , the magnitude of $\langle uv \rangle$ decreases with a . As was mentioned above, this is due to the gradual suppression of the vortex shedding. For most of the spin ratios, the maximum absolute values of the cross-stream Reynolds stresses are indicated on the down-wind side of the cylinder. The positive $\langle uv \rangle$ regions approach the surface for $a \geq 0.5$. One also notes an azimuthal displacement consistent with the direction of rotation. The same holds for the areas of negative Reynolds stresses. The iso-surfaces of these stresses shift closer to the wall for greater a , particularly for $a \geq 1$. The case $\alpha = 2$ exhibits different distributions of $\langle uv \rangle$. Indeed, the Reynolds shear stress peaks near the front stagnation point where the vorticity reaches its maximum. On the upper and downstream sides of the cylinder but close to its surface, u and v are negatively correlated and the shear stress reaches a minimum. At higher azimuthal angles ($\theta = 170^\circ$ – 180°) and a little further from the cylinder, positive values of the shear stress are observed but these are much lower than those on the upstream side of the cylinder.

4.3. Investigation of the wake stability as a function of the spin ratio

Flows past cylinders exhibit the well-known von-Karman vortex street for a wide range of Re number (Zdravkovich, 1997). At the present Re strong vortex shedding is observed accompanied by turbulent fluctuations. This wake instability has considerable effects on practical applications because the cylinder's loads fluctuate periodically with high amplitudes. Generally speaking, this

could be undesirable on many occasions and under certain circumstances could be catastrophic in structural applications.

Fig. 11 plots the lift and drag coefficient versus dimensionless time for various values of a . For $\alpha = 0$ and $\alpha = 0.5$, the lift coefficient oscillates periodically with high amplitudes over the whole time examined (at least 30 vortex shedding cycles). For $\alpha = 1$, the amplitudes are damped but they are still comparable with those plotted for the first 30 dimensionless time units. Careful inspection of even higher a , shows that the flow exhibits a transition period where the vortex shedding is quite strong. After that period, however, the amplitudes are damped and the vortex shedding is suppressed. During this time, the flow tends to be stable and statistically stationary. It is also expected (but not proved here) that for spin ratios greater than 2 the vortex shedding will be diminished.

The primary result deduced from Fig. 11 is that the flow becomes stable for $a \geq 1.5$ since the lift coefficient after a specific dimensionless time does not oscillate significantly, thus the forces on the cylinder stabilize. However, the value of a for wake stability has not been established. This value should lie in the range $1 < a \leq 1.5$. Three extra runs were thus performed to compute the approximate value of a that produces stability. Fig. 12 plots the history of the lift coefficient at $a = 1.2$ and $a = 1.3$. The variances are shown for all a greater or equal to 1 and additionally the variances scaled with the mean lift coefficient. The reason for computing the scaled variances is because the variance itself cannot provide an accurate description of the impact of the lift oscillations on the load stability, since the lift coefficient grows significantly with a and eventually the variance becomes higher as the mean value of C_l increases.

It is seen from the figure that at $a = 1.2$ the amplitudes of the fluctuation even for large dimensionless time are comparable with those in the transitional period; thus, the flow is still unstable. In contrast, for $a \geq 1.3$ the flow exhibits a different trend and the loads begin to be stabilized. Indeed, the variance is less than one third of the variance at $a = 1.2$ and reaches its minimum value of 0.014 for $a = 1.3$. The scaled variance reaches its minimum at

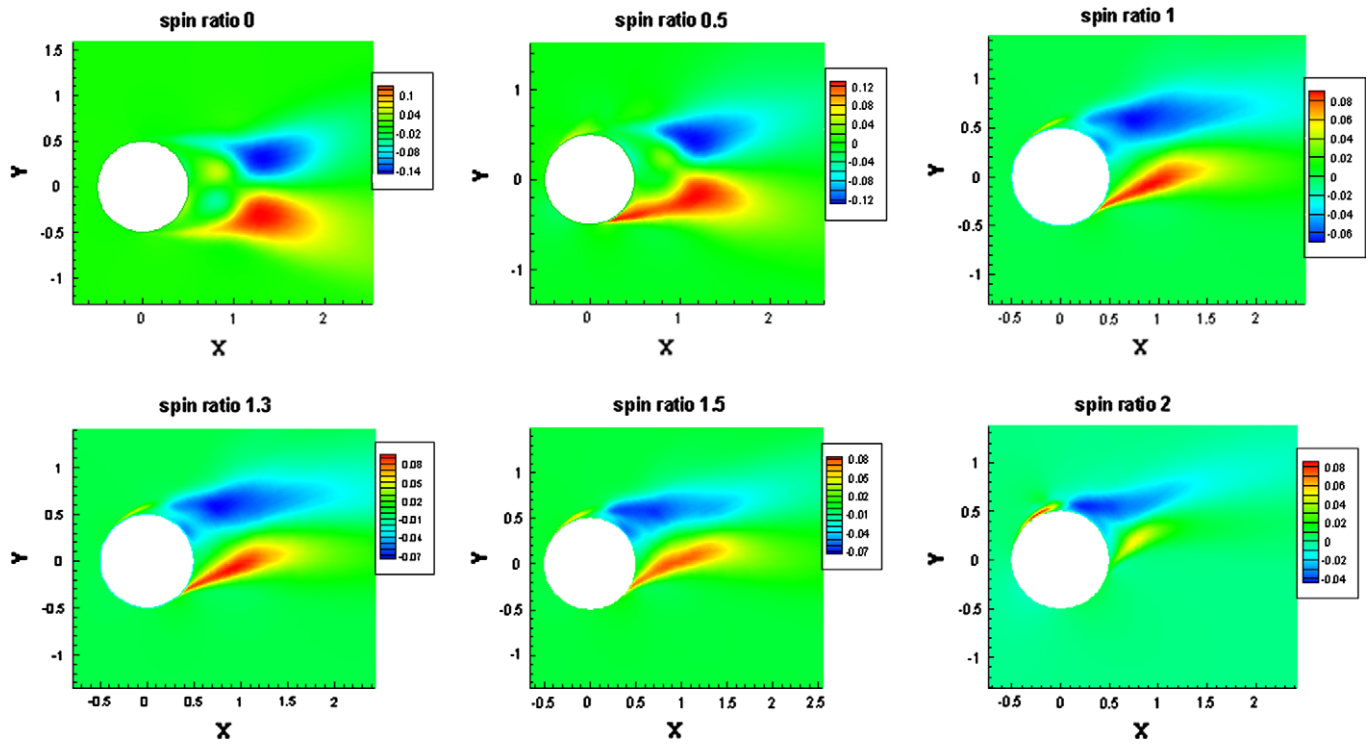


Fig. 10. Contours of the total resolved shear stress $\langle uv \rangle$ plotted in the near wake for the examined spin ratios.

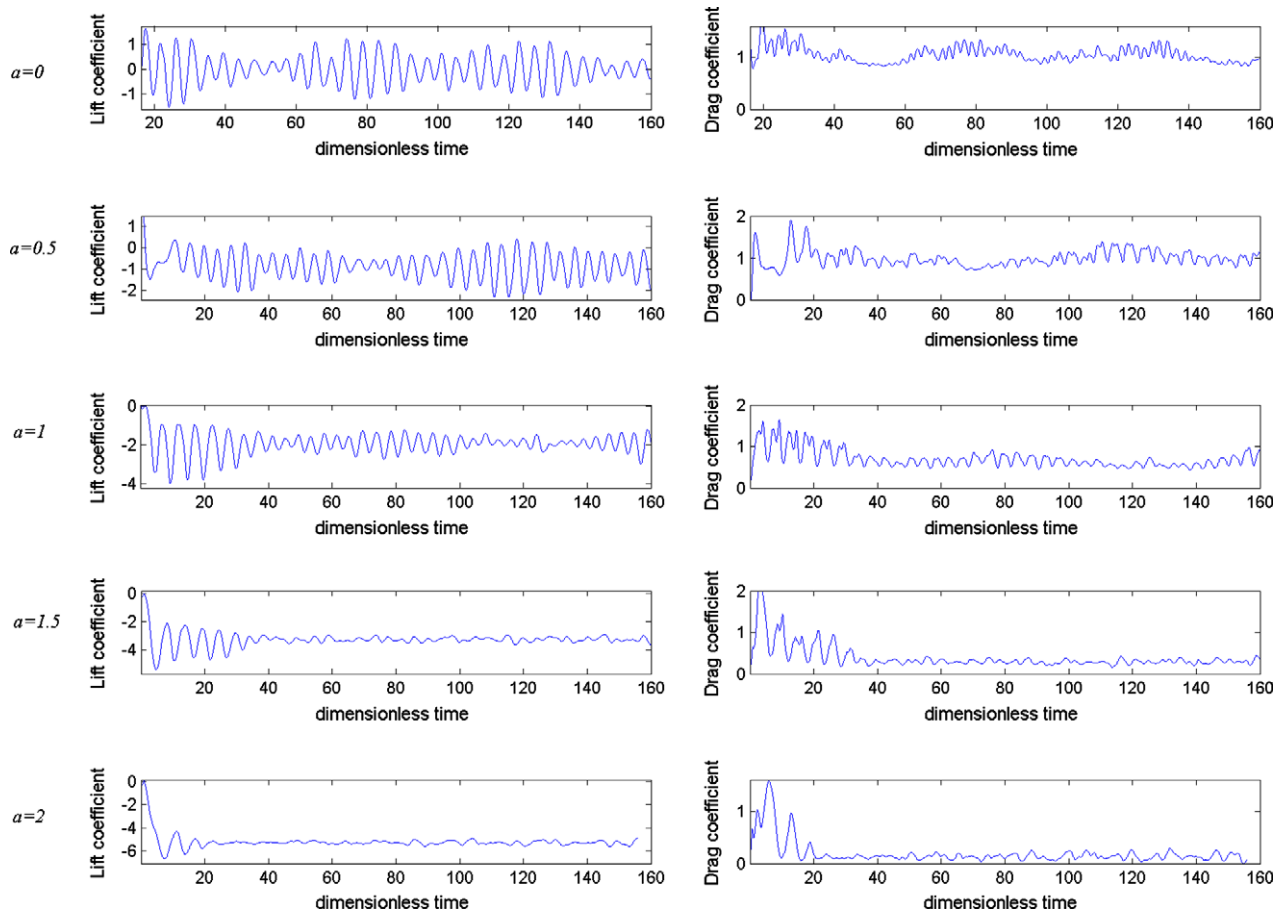


Fig. 11. Time histories of the lift and drag coefficient for various a .

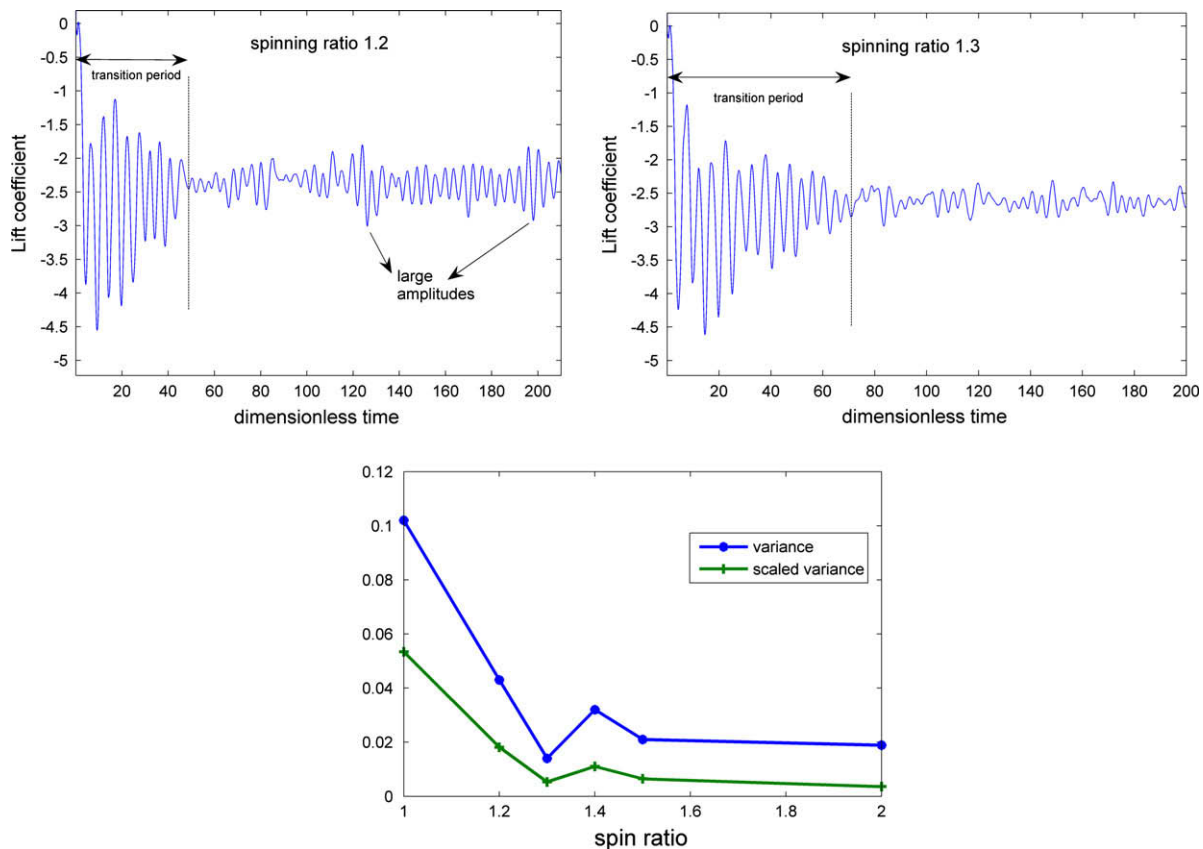


Fig. 12. Time history of C_l for $a = 1.2$, $a = 1.3$ and $a = 1.4$. The transition period is indicated.

$a = 2$, as expected, since the scaled fluctuations are very low at this spinning ratio. However, the flow cannot be characterized as stable since there are still high enough scaled amplitudes of C_l , as opposed to the case of $a = 2$. Nevertheless, the limit of $a = 1.3$ may be considered as the critical value of the spin ratio above which value the loads begin to become stable.

5. Conclusions

Uniform flow past a rotating cylinder is investigated with LES for spin ratios up to $a = 2$ and $Re = 140,000$. Since the Re number is so high, a well-validated SGS model has to be applied for the resolution of the smallest turbulent vortices. Based on the relevant studies of Breuer (1999, 2000), the Smagorinsky model was chosen because it produces equally accurate results as the dynamic model of Germano et al. (1991) at this Re number. The validation of the present code was quite successful for both the non-rotating and the rotating cases. Computations for the mean loads and the higher order moments agreed fairly well with other numerical simulations and experiments. An encouraging outcome from this study in terms of numerical modelling, is that accurate LES computations are feasible with the use of grids of medium resolution. Although wall functions are not used in the examined cases, the present grid is not extremely fine ($125 \times 125 \times 32$), but it was capable of resolving the viscous wall region at all spin ratios.

The flow patterns are strongly affected by a . As a increases, the downstream vortices are shifted to lower azimuthal angles. The lower shed vortex gradually shrinks and it eventually collapses. The upper one expands. The front stagnation point is shifted towards greater azimuthal angles as a rises. In contrast, the point of transition to turbulence is shifted upstream. The lift coefficient increases almost linearly with the spin ratio while the drag coefficient

decreases with the spin ratio and at $a = 2$ falls to the mean value of $C_d = 0.13$. At low a , the total resolved kinetic energy of the fluctuations is composed mainly of the variance of the periodic oscillations since the maximum values of k_f are indicated in the near wake of the cylinder. However, as a increases, turbulence produces strong fluctuations with amplitudes comparable with those of the periodic oscillations.

The wake up to $a = 1.2$ is unstable and the root mean square values of the loads on the cylinder are considerable. For $a \geq 1.3$, the loads are stabilized and the vortex shedding begins to be suppressed. In this range of a , the flow exhibits a transition ('transient') period with strong vortex shedding and then it becomes stable and almost statistically steady.

References

- Aoki, K., Ito, T., 2001. Flow characteristics around a rotating cylinder. In: Proceedings of the School of Engineering of Tokai University, vol. 26, pp. 29–34.
- Breuer, M., 1998. Numerical and modelling influences on large eddy simulations for the flow past a circular cylinder. *Int. J. Heat Fluid Flow* 19, 512–521.
- Breuer, M., 1999. Large eddy simulation of high Reynolds number circular cylinder flow. In: Biringen, S., Ors, H., Tezel, A., Ferziger, J.H. (Eds.), Proceedings of the Workshop on Industrial and Environmental Applications of Direct and Large Eddy Simulation. 5–7 August 1998, Bogazici University, Istanbul, Turkey, Lecture Notes in Physics, vol. 529. Springer, Berlin, pp. 176–189.
- Breuer, M., 2000. A challenging test case for large eddy simulation: high Reynolds number circular cylinder flow. *Int. J. Heat Fluid Flow* 21, 648–654.
- Cantwell, B., Coles, D., 1983. An experimental study on entrainment and transport in the turbulent near wake of a circular cylinder. *J. Fluid Mech.* 136, 321–374.
- Catalano, P., Wang, M., Iaccarino, G., Moin, P., 2003. Numerical simulation of the flow around a circular cylinder at high Reynolds numbers. *Int. J. Heat Fluid Flow* 24, 463–469.
- Elmiligui, A., Abdol-Hamid, K., Massey, J., Paul Pao, S. 2004. Numerical study of flow past a circular cylinder using RANS, Hybrid RANS/LES and PANS formulations. In: 22nd Applied Aerodynamics Conference and Exhibit, Rhode Island.
- Germano, M., Piomelli, U., Moin, P., Cabot, W.H., 1991. A dynamic subgrid scale eddy viscosity model. *Phys. Fluids A* 3 (7), 1760–1765.

- Hansen, R.P., Forsythe, J.R., 2003. Large and detached Eddy simulations of a circular cylinder using unstructured grids. *AIAA* 2003, 0775.
- Ingham, D.B., 1983. Steady flow past a rotating cylinder. *Comput. Fluids* 11, 351–366.
- Kang, S., Choi, H., Lee, S., 1999. Laminar flow past a rotating circular cylinder. *Phys. Fluids* 11, 3312–3320.
- Mittal, S., Kumar, B., 2003. Flow past a rotating cylinder. *J. Fluid Mech.* 476, 303–334.
- Padrino, J.C., Joseph, D.D., 2006. Numerical study of the steady-state uniform flow past a rotating cylinder. *J. Fluid Mech.* 557, 191–223.
- Smagorinsky, J., 1963. General circulation experiments with the primitive equations. I. The basic experiment. *Mon. Weather Rev.* 91, 99–164.
- Stojkovic, D., Breuer, M., Durst, F., 2002. Effect of high rotation rates on the laminar flow around a circular cylinder. *Phys. Fluids* 14 (9), 3160–3178.
- Stojkovic, D., Schon, P., Breuer, M., Durst, F., 2003. On the new vortex shedding mode past a rotating circular cylinder. *Phys. Fluids* 15 (5), 1257–1260.
- Travin, A., Shur, M.L., Strelets, M., Spalart, P.R., 2000. Detached Eddy simulation past a circular cylinder. *Flow Turbul. Combust.* 63, 193–313.
- Wieselsberger, C., Betz, A., Prandtl, L., 1923. *Versuche uber den Widerstandgerundeter und kantiger Korper*. Ergebnisse AVA Gottingen. pp. 22–32.
- Zdravkovich, M.M., 1997. *Flow around Circular Cylinders. Fundamentals*, vol. 1. Oxford University Press, New York.